

## The $w \rightarrow z$ region of $N_2$

Where  $f f U(w)U(z) = N_c U$  actually fires — and whether it can make a divergence

$U$  adjoint, real orthogonal;  $f^{abc}$  totally antisymmetric;  $N_c = 3$  throughout. Ket =  $f^{cbd} U^{ac}(w) \rho^d(x) - f^{acd} U^{bc}(z) U^{de}(x) \rho^e(x)$ , free indices  $a$  (observed gluon, at  $w$ ) and  $b$  (unobserved gluon, at  $z$ ).

**Short answer: no divergence, and the identity does not fire at  $w \rightarrow z$ .** The collapse  $f f U U \rightarrow N_c$  needs the two Wilson lines to sit at the same point *and* to be contracted through a common summed index, so that  $\sum_a U^{ac'} U^{ac} = \delta^{c'e}$  fires first. In the cross terms  $U(w)$  carries the **observed** gluon's colour  $a$  and  $U(z)$  carries the **unobserved** gluon's colour  $b$ : two different free indices, one on each  $f$ . Setting  $w = z$  makes the matrices equal but leaves the index structure alone — no  $\delta$ , no  $N_c$  (§3, numerically confirmed). The identity *does* fire in two other places (§2), and in both the transverse integrand is convergent anyway. And in any case (§4) the integrand at  $w \rightarrow z$  is  $\rho^{-1}$ , not  $\rho^{-2}$ , with the full colour factor in place — and colour is a bounded contraction of orthogonal matrices, so it can only make a region **more** convergent, never less.

### 1. The identity, verified

Because  $U(z)$  is an adjoint rotation and  $f$  an invariant tensor,  $f^{a'b'c'} U^{aa'} U^{bb'} U^{cc'} = f^{abc}$ ; contracting one leg with  $U^T$  gives the form you want,

$$f^{abc} U^{cd}(z) = U^{aa'}(z) U^{bb'}(z) f^{a'b'd}, \quad f^{acd} f^{bcd} = N_c \delta^{ab}.$$

Numerically:  $|U^T U - 1|_{\max} = 4 \cdot 10^{-16}$ , first identity  $3 \cdot 10^{-16}$ , second  $1 \cdot 10^{-15}$ . Both are needed: the second is what actually produces the  $N_c$ , and it needs a  $\delta$  to have appeared on a pair of  $f$  legs first.

### 2. Where the collapse really happens in this row

#### (a) The $(2\rho) \times (2\rho)$ colour factor — *unconditionally*

Bra and ket both carry  $U(\cdot)$  at the **same** point  $z$ , sharing the summed index  $b$ :

$$\sum_b U^{bc'}(z) U^{bc}(z) = \delta^{c'c} \implies \sum_{a,c'} f^{ac'd'} f^{ac'd} = N_c \delta^{d'd} \implies \boxed{\mathcal{C}_{22}^{e'e} = N_c [U(x') U(x)]^{e'e}}$$

verified to  $8 \cdot 10^{-15}$ . This is your identity, and it is worth noticing: **the  $(2) \times (2)$  colour factor carries no  $z$  and no  $w$  dependence at all**, so it pulls straight out of every transverse integral. (That is the piece still missing from your expression — see §1.3 of the previous note.)

#### (b) The $(1\rho) \times (1\rho)$ colour factor at $w' \rightarrow w$

Here the shared summed index is  $a$ :

$$\sum_a U^{ac'}(w') U^{ac}(w) \xrightarrow{w' \rightarrow w} \delta^{c'c} \implies \mathcal{C}_{11}^{d'd} |_{w'=w} = N_c \delta^{d'd}$$

verified to  $2 \cdot 10^{-15}$ , against  $\|\mathcal{C}_{11}\|_{\max} = 0.94$  at generic  $w'$ . So there is an  $N_c$  enhancement — but at  $w' = w$ , not at  $w = z$ .

### 3. Why it does *not* fire at $w \rightarrow z$

The cross terms are

$$\Sigma^{d'd} = \sum_{a,b,c,c'} f^{ac'd'} U^{bc'}(z) f^{cbd} U^{ac}(w).$$

The two Wilson lines sit on **different** free indices:  $U^{ac}(w)$  on the observed gluon's colour  $a$ ,  $U^{bc'}(z)$  on the unobserved gluon's colour  $b$ ; and  $a$  and  $b$  are summed independently, one against each  $f$ . Putting  $w = z$  makes  $U(w) = U(z)$  as matrices, but it does not bring  $a$  and  $b$  together, so  $\sum_a U^{ac'} U^{ac}$  never appears and there is no  $\delta$  to feed  $f^{acd} f^{acd'} = N_c \delta^{dd'}$ .

Numerically:

| quantity                                                   | value                    |
|------------------------------------------------------------|--------------------------|
| $\ \Sigma(w=z)\ _{\max}$                                   | 0.973551                 |
| $\ \Sigma(w)\ _{\max}$ , 500 random $w$ — min / mean / max | 0.5945 / 1.0100 / 1.8493 |
| $\ \Sigma(w=z) - N_c \delta^{d,d}\ $                       | 3.97                     |
| $\ \Sigma(w=z) \mp N_c U(z)\ $                             | 2.52 / 2.39              |
| $\ \Sigma(w=z) \mp \frac{N_c}{2} U(z)\ $                   | 1.41 / 1.54              |

No  $N_c$  form fits, and  $\Sigma(w=z)$  sits comfortably *inside* the generic range. The point  $w = z$  is colour-wise completely unremarkable.

#### 4. And in any case the transverse integrand forbids a log there

Scanning the **full** integrand — transverse structure  $\times$  phase  $\times$  the complete colour factor with non-trivial  $U(x), U(x'), U(z), U(w), U(w')$  — as  $\rho = |w - z| \rightarrow 0$ :

| $\rho$    | $\langle  \text{full}  \rangle$ | $\langle  \text{full}  \rangle \rho$ | $ \langle \text{full} \rangle $ <b>signed</b> |
|-----------|---------------------------------|--------------------------------------|-----------------------------------------------|
| $10^{-1}$ | $2.580534 \cdot 10^1$           | 2.580533612                          | 1.398661482                                   |
| $10^{-2}$ | $2.579680 \cdot 10^2$           | 2.579679634                          | 1.399894333                                   |
| $10^{-3}$ | $2.579671 \cdot 10^3$           | 2.579671403                          | 1.399906683                                   |
| $10^{-4}$ | $2.579671 \cdot 10^4$           | 2.579671320                          | 1.399906807                                   |
| $10^{-5}$ | $2.579671 \cdot 10^5$           | 2.579671320                          | 1.399906888                                   |
| $10^{-6}$ | $2.579671 \cdot 10^6$           | 2.579671320                          | 1.399916580                                   |

$\langle |\text{full}| \rangle \rho$  is flat to ten digits: the integrand is  $\rho^{-1}$ , so against  $d^2 \rho \sim \rho d\rho$  it is  $\int d\rho$  — **linearly convergent, no log**. A logarithm needs  $\rho^{-2}$ . Better still, the *signed* angular average tends to a **constant**, not to  $\rho^{-1}$ : every  $\rho^{-1}$  piece of  $\mathcal{W}^{ji}$  is odd in  $(z - w)$  and averages away, so the true behaviour is  $O(1)$  and the region is softer even than the modulus bound suggests.

**The general point.** A colour factor cannot manufacture a divergence. Every one of them here is a finite contraction of  $f$ 's with *orthogonal* matrices, so  $|\mathcal{C}|$  is bounded uniformly in all transverse arguments. Colour can only **cancel** a would-be divergence (by vanishing in the limit); it can never create one. The power counting is therefore settled by the transverse structure alone — which is why the Wilson-line-free scan of the previous note was already conclusive.

#### 5. The region where the identity does fire is regular

At  $w' \rightarrow w$ , where  $\mathcal{C}_{11} \rightarrow N_c \delta^{d,d}$ :

| $\rho =  w' - w $ | $\langle  \text{full}  \rangle$ | $\langle  \text{full}  \rangle \rho$ |
|-------------------|---------------------------------|--------------------------------------|
| $10^{-1}$         | 6.467434                        | 0.646743                             |
| $10^{-2}$         | 6.474616                        | 0.064746                             |
| $10^{-3}$         | 6.474687                        | 0.006475                             |
| $10^{-4}$         | 6.474688                        | 0.000647                             |
| $10^{-5}$         | 6.474688                        | 0.000065                             |

$\langle |\text{full}| \rangle$  itself goes to a constant: the integrand is **regular** at  $w' = w$ . So the  $N_c$  collapse multiplies a perfectly convergent region and changes nothing. (It is also the  $k_\perp$ -integrated corner, oscillation-suppressed by  $e^{-ik_\perp(w'-w)}$  for any  $k_\perp \neq 0$ .)

## 6. Conclusion

- **No divergence at  $w \rightarrow z$ .** Transverse behaviour  $\rho^{-1}$  (signed:  $O(1)$ ), measure  $\rho d\rho$ : convergent. This confirms the earlier conclusion with the colour structure fully included rather than bounded away.
- **The identity does not apply at  $w \rightarrow z$ ,** because  $U(w)$  and  $U(z)$  hang off different final-state colour indices. It applies at  $w' \rightarrow w$  in  $\mathcal{C}_{11}$ , and unconditionally in  $\mathcal{C}_{22}$ .
- **The useful by-product:**  $\mathcal{C}_{22} = N_c[U(x')U(x)]^{e'e}$  exactly, with no  $z$  or  $w$  dependence — so when you add the missing  $(2\rho) \times (2\rho)$  piece, its colour factor factorises out of the whole transverse integral. That should make it the easiest of the three pieces to do.

Backed by `src/row17_wz.py`; raw output appended to `RESULTS_row17.txt`.